

Mikayel Harutyunyan's Report File

DS223 - Homework 1

2. Identify a similar innovation from the past.

The Jackery Solar Roof is closely related to the broader innovation of solar photovoltaic (PV) systems, which generate electricity directly from sunlight. Like traditional rooftop solar panels, the Jackery Solar Roof uses photovoltaic cells to convert solar energy into usable electricity for households. The key difference is in the design: instead of visible solar panels mounted on top of the roof, Jackery integrates the solar technology directly into curved roof tiles, allowing the system to blend more naturally with the appearance of the house.

A similar idea has already appeared in the market with products such as the Tesla Solar Roof, which also integrates photovoltaic cells into roof tiles. These innovations show how solar technology has evolved from purely functional panels to solutions that also consider aesthetics and home design. Because the underlying technology remains the same as conventional solar PV systems, the historical diffusion of solar photovoltaic installations provides a reasonable proxy for understanding how innovations like the Jackery Solar Roof may spread in the market.

3. Find historical data

Historical data for the analysis uses global cumulative solar photovoltaic (PV) capacity. The dataset reports the total installed solar PV capacity worldwide from 2000 to 2024, capturing the growth and adoption of solar panel technology over time. Since the Jackery Solar Roof performs

the same core function, which is generating electricity from sunlight, this time series provides a suitable proxy for estimating the Bass diffusion model parameters.

Retrieved from Statista: [Global cumulative solar PV capacity \(2000–2024\)](#)

4. Estimate Bass Model parameters

Bass parameters were estimated using nonlinear least squares and the diffusion package to ensure robustness. The results indicate that the technology is still in a very rapid growth phase. The Bass diffusion model typically captures an S-shaped adoption curve, but the data does not yet show this pattern. Instead, annual sales increased from 327 MW to 576,000 MW over 24 years, suggesting continued strong growth. Since the slowdown phase has not yet appeared, the models must estimate the future market ceiling, which explains the differences between the two estimation methods.

I applied the two estimation approaches presented in the lecture slides, with some minor adjustments to the first method to improve the parameter estimation.

For the NLS estimation, I experimented with many different initial parameter values to help the optimization converge. Some of the final parameters may look somewhat extreme, but they were the ones that produced the most stable estimation and the most reasonable forecast curve.

The *NLS estimation (Method 1)* results in $p \approx 0.00012$, $q \approx 0.164$, and $M \approx 31$ million MW. In my final plot, the fitted curve (colored green) follows the observed data points fairly close. The relatively high q compared to p indicates that imitation effects dominate adoption, suggesting that diffusion appears to be strongly influenced by social diffusion effects, such as recommendations and conversations among people.

The *diffusion package estimation (Method 2)* produces $p \approx 0.0015$, $q \approx 0.134$, and $M \approx 7.15$ million MW. This model generates a more moderate adoption path, with slower growth and a lower estimated market potential.

In addition, the NLS estimation did not fully converge. This could be caused by the small data and the early stage of it, as it has not yet reached the maturity phase. Without observing a peak or saturation, the model struggles to determine the eventual market size precisely.

Another challenge is the large scale of the data. The values range from a few hundred megawatts to several hundred thousand megawatts, which can make numerical optimization more difficult and reduce the precision of the parameter estimates.

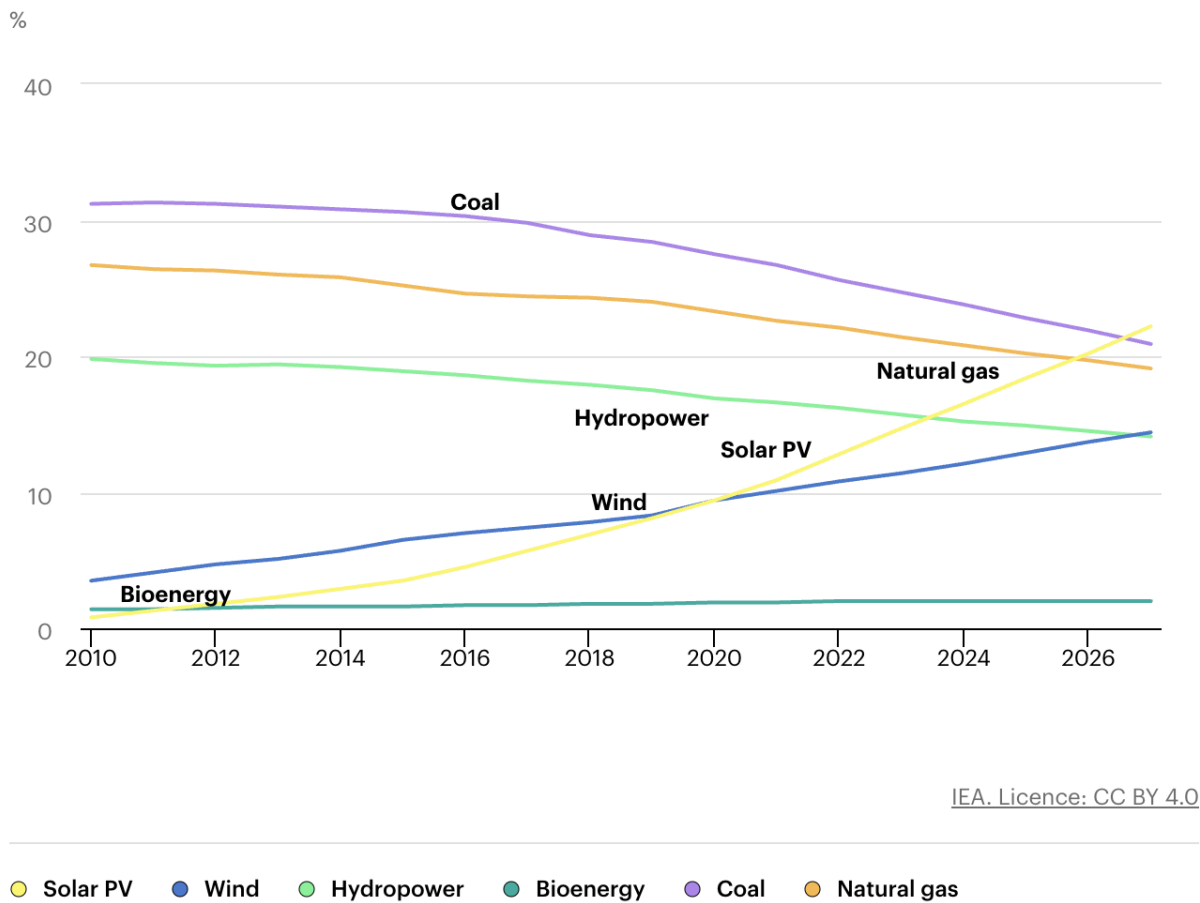
So I think the NLS model provides a more reasonable representation of the current diffusion pattern, even though the long-term market potential remains uncertain.

5. Predict the diffusion of the innovation selected in step

The diffusion path was projected through 2035 and shows a continued increase in annual solar PV adoption. This aligns with statistics from the International Energy Agency, which reports that solar PV is currently the fastest-growing source of electricity generation worldwide. Global installed solar capacity has already surpassed **2 terawatts**, and the agency projects that solar will become the largest source of global power capacity by **2027**. These trends support the model's relatively high imitation coefficient, suggesting that adoption is increasingly driven by market and social dynamics as solar technology becomes more affordable and widely adopted.

Share of cumulative power capacity by technology, 2010-2027

Open 



Retrieved from IEA: [International Energy Agency Solar PV statistics](#)

6. Choose a scope (Country: Netherlands)

In this phase, I focused on the Netherlands to examine how residential solar PV systems are spreading within a specific market. From 2019 to 2024, installed systems increased rapidly from about 960,000 to nearly 2.8 million, showing strong growth. The Bass diffusion model suggests adoption is mainly driven by imitation ($q = 0.95$), meaning households are strongly influenced by others installing solar, while the role of innovators is smaller ($p = 0.046$).

The model estimates the peak adoption period around 2021, but the data suggests the fastest growth occurred closer to 2022–2023. This difference likely results from the small dataset, which causes the model to smooth the trend and predict an earlier peak. In reality, the solar market in the Netherlands is still growing rapidly, likely supported by various factors, thus a simple Bass model is not enough for precise predictions.

Retrieved from Statista: [Residential Solar PV Systems in the Netherlands \(2019-2024\)](#)

7. Estimate the number of adopters by period.

Using the estimated Bass model parameters, I approximated the potential number of adopters over time. The Bass model estimated the market potential parameter m , but since the dataset is small and the model already underestimates the observed adoption, I used Fermi's logic to validate the market size.

First, I estimated the potential number of households that could install residential solar PV systems in the Netherlands.

1) Approximate Population of the Netherlands: 18,000,000 people (According to [European Union](#))

2) Average household size: 2.2 people per household

3) Estimated number of households: $18,000,000 / 2.2 \approx 8,180,000$

4) Share of households with suitable rooftops

Not all households can install solar panels because of apartment living, shading, or unsuitable roofs.

Assuming roughly 40% of households have suitable rooftops:

$$8,180,000 \times 0.40 = 3,272,000$$

5) Estimated market potential

$M \approx 3.27$ million households

This estimate is consistent with the observed data, which shows that by 2024 approximately **2.79 million** systems had already been installed. Given the Fermi estimate of $m(F) = 3.27$ million potential households, the market has not yet fully saturated but is clearly approaching its upper bound. Looking at the graph, with $m(F)$ the projection starts relatively slowly and then begins to slightly overestimate adoption, but it still captures the overall market pattern better than the market potential estimated by the Bass diffusion model.

Based on the estimated Bass model parameters, adoption begins slowly with a small group of innovators, then accelerates as imitation effects become dominant, and eventually slows as the market approaches saturation. Again, as the imitation coefficient q is much larger than the innovation coefficient p , the diffusion process is primarily driven by social influence, such as social influence and market trends before deciding to install solar panels themselves.

References:

Netherlands. European Union. (2025).

https://european-union.europa.eu/principles-countries-history/eu-countries/netherlands_en

Lucía Fernández, & 28, N. (2025a, November 28). *Netherlands: Number of residential solar PV systems 2024*. Statista.

<https://www.statista.com/statistics/1616980/number-residential-solar-pv-systems-netherlands>

So, W. (2026, January 20). *Global cumulative installed solar PV capacity 2024*. Statista.

<https://www.statista.com/statistics/280220/global-cumulative-installed-solar-pv-capacity>

Solar PV. International Energy Agency (IEA). (n.d.).

<https://www.iea.org/energy-system/renewables/solar-pv>